EUNOIA JUNIOR COLLEGE
JC2 Preliminary Examination 2022
General Certificate of Education Advanced Level
Higher 2

H2 Biology

9744/01

Paper 1 Multiple Choice

21 September 2022

60 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use paper clips, glue or correction fluid.
Write your name, civics group and registration number on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

This document consists of **20** printed pages.

Answer **all** questions.

1 Tests on four samples from a mixture of biological molecules gave the results shown in the table.

test	boiled with excess Benedict's solution	boiled with excess Benedict's solution after acid hydrolysis and neutralisation	biuret reagent	iodine solution
result	blue	red	purple	yellow

Which biological molecules were in the mixture?

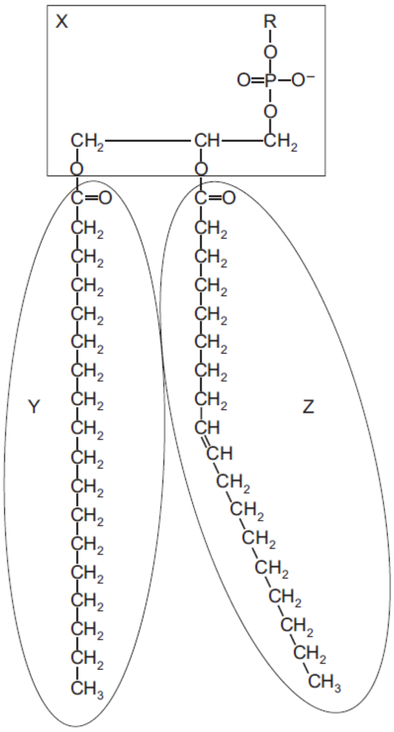
A reducing sugar and protein

B reducing sugar, non-reducing sugar and starch

C non-reducing sugar and protein

D non-reducing sugar and starch only

2 The diagram shows a phospholipid molecule divided into three regions, X, Y and Z. R, in region X, represents a range of possible chemical groups.



Regions X, Y and Z affect the properties of cell surface membranes in different ways.

Which row shows the effect of each region on the properties of a cell surface membrane?

	increases permeability of hydrophobic region	repels polar molecules	attracts water molecules
A	X	X	Y and Z
B	Y	Y and Z	X
C	Y and Z	X	Y and Z
D	Z	Y and Z	X

3 Which statements could be used to describe enzyme molecules **and** antibody molecules?

1 Hydrogen bonds stabilise the structure of the protein and are important for it to function efficiently.

2 Hydrophilic R groups point in towards the centre of the molecule and cause it to curl into a spherical shape.

3 The tertiary structure of the protein molecule plays an important role in the functioning of the protein.

A 1 and 2 only

B 1 and 3 only

C 2 and 3 only

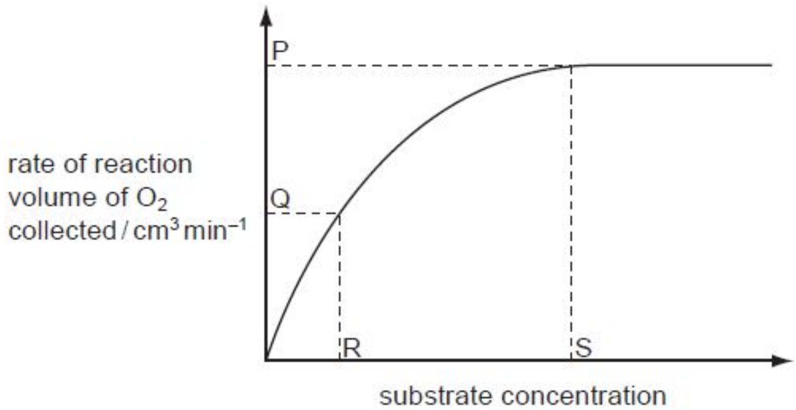
D 1, 2 and 3

4 Liver tissue produces an enzyme called catalase which breaks down hydrogen peroxide into water and oxygen.

$$2H_2O_2 \rightarrow 2H_2O + O_2$$

The rate of this reaction can be determined by measuring the volume of oxygen produced in a given length of time.

Students added small cubes of fresh liver tissue to a range of hydrogen peroxide solutions and measured the volumes of oxygen produced. Their data were used to produce the graph showing how changing the concentration of hydrogen peroxide affected the rate of oxygen production.



Which statement is correct?

1 At P, the rate of reaction is limited by the concentration of enzyme.

2 At Q, all of the enzyme active sites are occupied by substrate molecules.

3 At Q, the rate of reaction is limited by the concentration of the substrate.

4 R represents K_m where the reaction rate = $V_{max} / 2$.

5 At S, all of the enzyme active sites are occupied by substrate molecules.

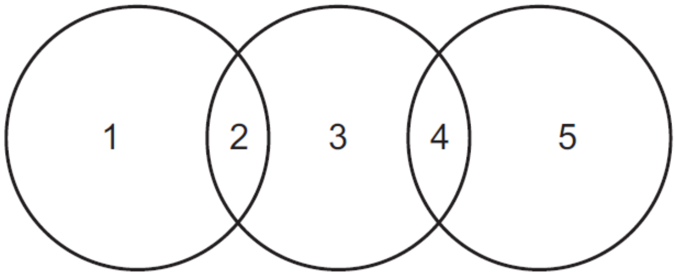
A 2 and 3 only

B 2 and 5 only

C 1, 4 and 5 only

D 1, 3, 4 and 5 only

5 The diagram shows the relationship between various cells and their components.



Which row is correct?

	1	2	3	4	5
A	80S ribosome	eukaryotic cell	mitochondrion	70S ribosome	prokaryotic cell
B	chloroplast	plant cell	cell wall	prokaryotic cell	80S ribosome
C	circular DNA	nucleus	eukaryotic cell	mitochondrion	70S ribosome
D	prokaryotic cell	circular DNA	chloroplast	membrane bound	70S ribosome

6 The eyepiece lens of a microscope can be fitted with an eyepiece graticule.

Which of these statements about eyepiece graticules are correct?

1 They measure the actual length of cells in micrometres.
2 They help biologists to draw cells with correct proportions.
3 They change in size when the objective lens is changed from x10 to x40.

A 1 only
B 2 only
C 1 and 3 only
D 1, 2 and 3

7 Liver cells contain vesicles that have proteins in their membranes which are specific for the transport of glucose.

When these cells need to take up glucose, the vesicles fuse with the cell surface membrane.

How does the uptake of glucose occur?

A Exocytosis
B Diffusion
C Endocytosis
D Facilitated diffusion

8 A piece of a DNA molecule contains 84 base pairs. The table shows the number of adenine and cytosine bases in one or both of the DNA strands in this piece of DNA molecule.

base	strand 1	strand 2
adenine	28	23
cytosine	15	

How many guanine bases are present in this piece of DNA molecule?

A 18 **B** 33 **C** 36 **D** 41

Strand 1 has 23 T (since there are 23 A on Strand 2).
Strand 1 has 28 + 23 + 15 = 66 (A+T+C). Therefore, it has 84 - 66 = 18 G.
Strand 2 has 15 G (since there are 15 C on Strand 1).
Strand 1 and 2 has 18 + 15 = 33 G.

9 A polypeptide has the amino acid sequence glycine – arginine – lysine – serine.

The table below gives possible tRNA anticodons for each amino acid.

amino acid	possible tRNA anticodons	
Arginine	3' UCC 5'	3' GCG 5'
Glycine	3' CCA 5'	3' CCU 5'
Lysine	3' UUC 5'	3' UUU 5'
Serine	3' AGG 5'	3' UCG 5'

Which sequence of bases on DNA would code for the polypeptide?

A 3' CGA CGC AAG AGC 5'

B 3' CCT TCC TTC TCG 5'

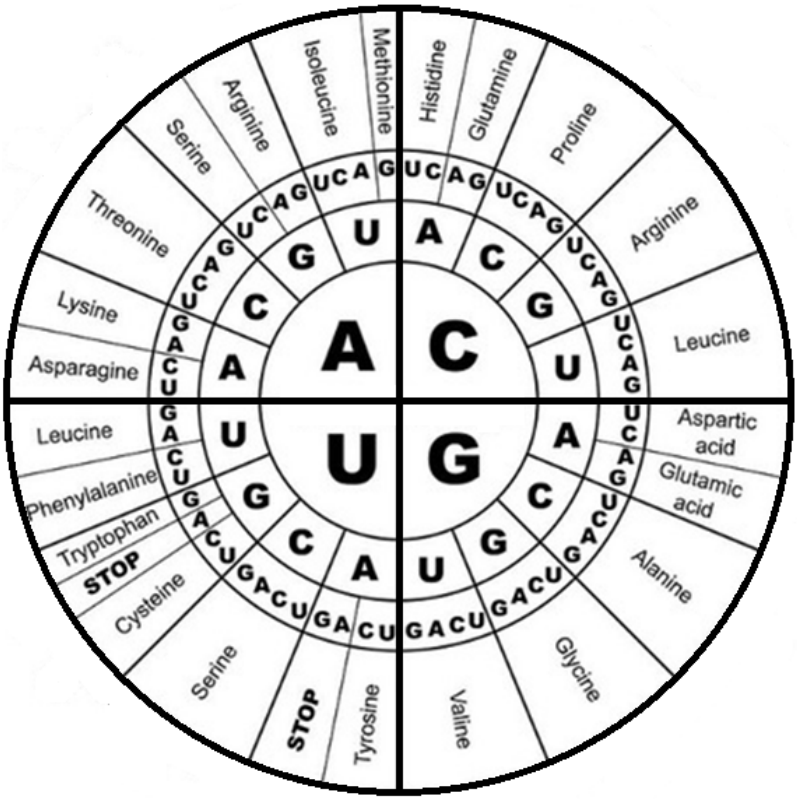
C 5' GGA AGG AAA AGC 3'

D 5' GGT TGG TTG TGC 3'

Appropriate anticodons are complementary to mRNA codons which in turn are complementary to the DNA template. The anticodons thus have **similar sequence** to the DNA template strand triplet codes for the given sequence of amino acids. For glycine – arginine – lysine – serine, we have for glycine, either 3' CCA 5' or 3' CCU 5' anticodons, this corresponds to the starting triplet of option B, 3' CCT 5', further inspection would confirm the sequence given in option B to be correct for rest of the triplet codes. Note the direction of transcription of the DNA template strand (3' to 5') and the direction of translation of the mRNA strand (5' to 3').

DNA triplet codes:	3' CCT	TCC	TTC	TCG 5'
mRNA codons:	5' GGA	AGG	AAG	AGC 3'
Anticodons:	3' CCU 5'	3' UCC 5'	3' UUC 5'	3'UCG 5'

10 The mRNA codons for amino acids are shown below.



A mutagen causes the adenine in DNA to pair with cytosine during transcription.

Which tripeptide will be synthesised when the template DNA sequence **5’ TAACTGCCA 3’** is used in protein synthesis in the presence of this mutagen?

A proline-valine-proline

B arginine-glutamine-proline

C tryptophan-glutamine-leucine

D threonine-valine-asparagine

Template DNA: 3’ **A**CC GTC **AA**T 5’
mRNA: 5’ **C**GG CAG **CC**A 3’

11 A student observed the cells in the growing region (meristem) of an onion root and obtained the data shown.

stage	number of cells
interphase	886
prophase	73
metaphase	16
anaphase	14
telophase	11

Which percentage of cells contains chromosomes that appear as two chromatids?

A 7.3

B 8.9

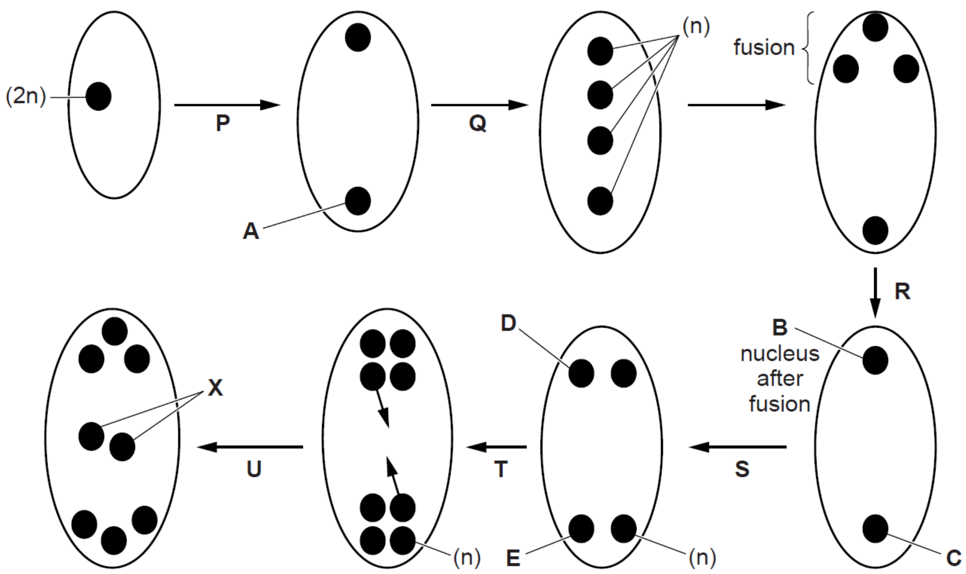
C 95.9

D 97.5

Total number of cells = 886 + 73 + 16 + 14 + 11 = 1000
Stages where cells contain **chromosomes** that appear as **two chromatids**: Prophase & Metaphase
Percentage of cells that contain **chromosomes** that appear as **two chromatids** = (73 + 16) / 1000 = 8.9%

12 The development of the embryo sac in flowering plants involves both mitosis and meiosis. Details of this development can vary in different plants.

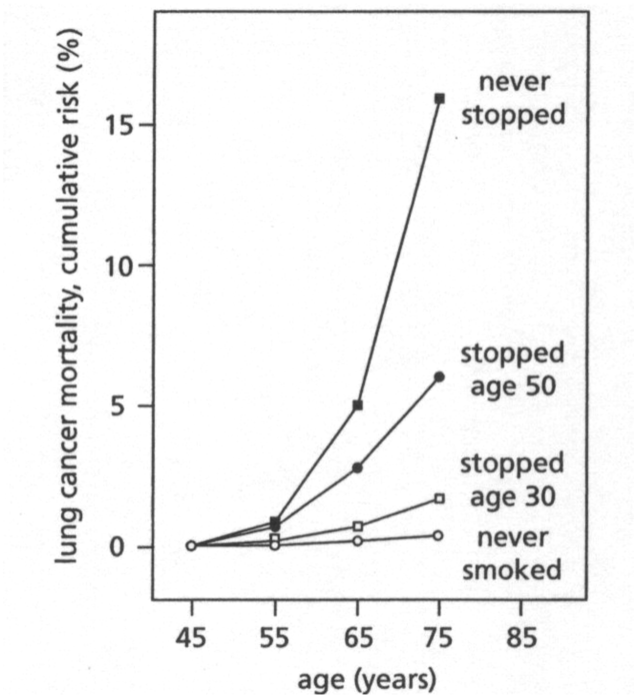
The diagrams summarise the development of the egg cell within the embryo sac of *Lilium* sp. Some of the nuclei have been labelled to indicate the ploidy: n = haploid; 2n = diploid.



Which stage, from **P**, **Q**, **R**, and **T**, represents meiosis II?

A P **B** Q **C** R **D** T

13 Mortality due to lung cancer was followed in groups of males in the United Kingdom for 50 years. The cumulative risk of dying from lung cancer as a function of age and smoking habits for four groups of males is shown in the figure.



Which of the following best explains the trends observed in the graph?

I	The slower the rate of accumulation of mutations, the lower the cumulative risk.
II	When an individual stops smoking, he will undergo a decreased rate of mutation accumulation.
III	The presence of cigarette smoke will cause the accumulation of mutations to occur at an increased rate.
IV	There is little risk of a non-smoker dying of lung cancer.

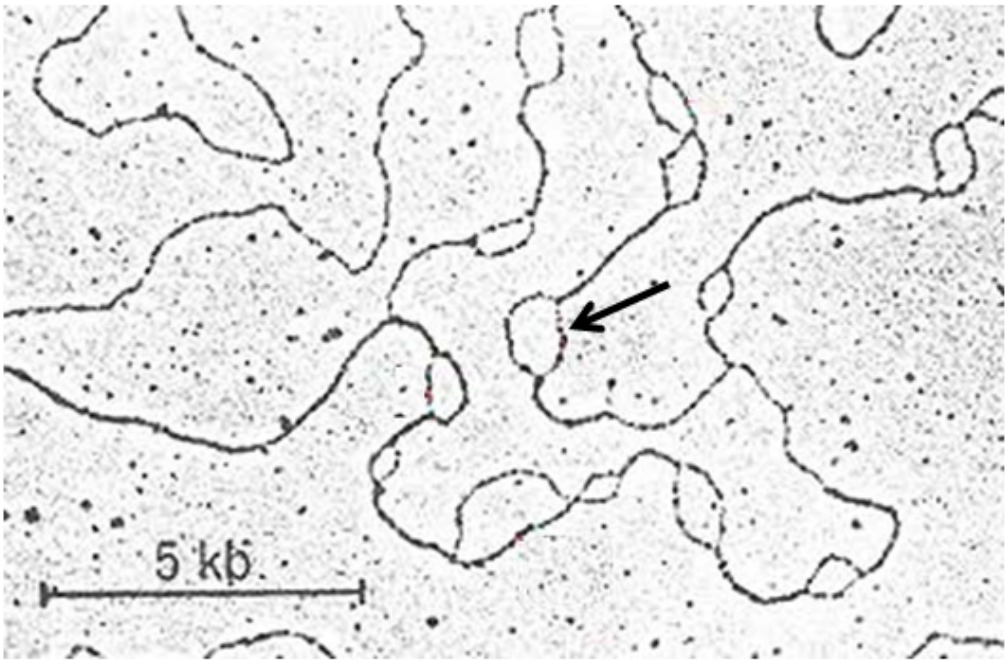
A **I** and **IV** only

B **II** and **III** only

C **III** and **IV** only

D All of the above

14 The figure below shows structures that are formed along a DNA molecule of a eukaryotic cell during S phase of the cell cycle.



Which statement(s) correctly describe the labelled structure and the process that is taking place?

1 Many such structures can also be found in prokaryotic DNA.

2 DNA replication requires primase to synthesize primers to provide free 3' OH for the elongation of daughter strands by RNA polymerase.

3 Eukaryotic chromosomes face the end-replication problem as they are linear. Prokaryotic chromosomes do not have the end-replication problem as they are circular.

4 The end-replication problem occurs only on the lagging strands.

A 1 and 2 only

B 1 and 3 only

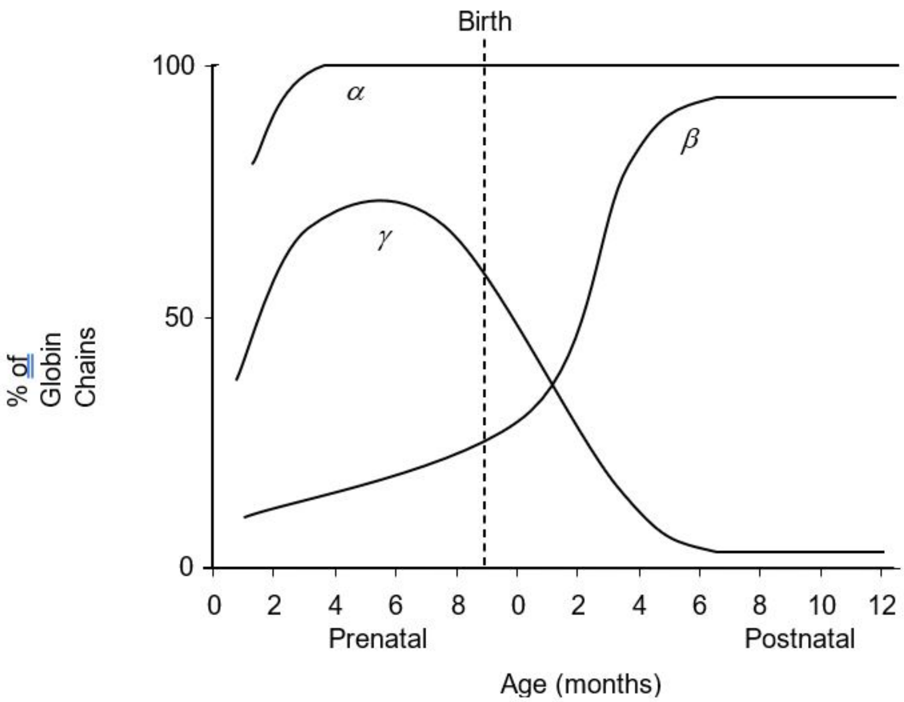
C 3 only

D 3 and 4 only

Explanation:
The TEM shows the replication bubbles along the DNA of a cultured Chinese hamster cell. In each linear chromosome of eukaryotes, DNA replication begins when replication bubbles form at many sites along the giant DNA molecule. The bubble expands as replication proceeds in both directions. Eventually the bubble fuses and synthesis of the daughter strands is complete.

15 The globin gene family in humans consists of the α, β and γ genes. These genes code for the globin chains that make up haemoglobin and are expressed at different levels during different developmental stages.

The graph shows the expression of the various globin chains during the prenatal (fetal) and postnatal (after birth) periods.



Which of the following cannot account for the differences in the levels of expression of globin chains?

A Methyl groups are added to regulatory sequences of γ-globin genes during the postnatal period, allowing for some proteins to bind.

B Alternative splicing occurs in the mature mRNA of the α-globin and β-globin genes, resulting in differences in the rate of expression of globin chains during the prenatal period.

C A growth factor triggers the expression of a transcription factor that increases the rate of β-globin gene expression during the postnatal period.

D The shortening of poly(A) tail in the mRNA of γ -globin genes reduces its stability, resulting in a decrease in the rate of expression of γ-globin chains during the postnatal period.

Explanation:
Alternative splicing does not change the rate of gene expression

16 One hypothesis about the origin of viruses is that they existed before cellular life forms, later evolving into parasites of cellular organisms. Groups with an ancient, common origin tend to share conserved sequences of DNA or RNA.

Which observation about virus genomes supports the view that viruses existed before cellular life forms and only much later evolved into parasites?

A All viruses have genes that code for capsid components.

B Introns of eukaryotic genes have common features with viral genomes.

C Large virus genomes have genes that originate in host cells.

D Viral genomes have conserved genes not found in any other genomes.

Explanation

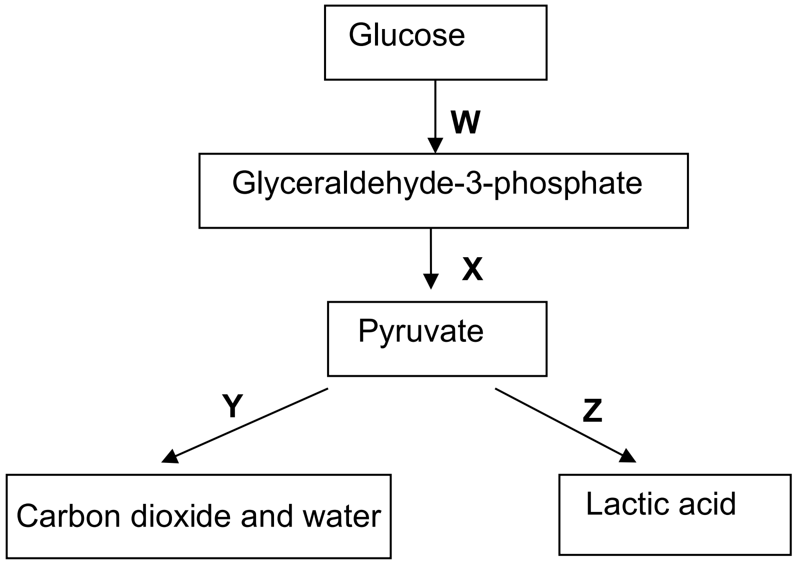
The observation that all viruses share conserved genes that are not found in any other genomes suggests that viruses existed before other life forms evolved.

17 The diagram shows an electron micrograph of a bacterial cell.

Which of the following correctly identifies the functions of structures **W**, **X**, **Y** and **Z**?

	W	X	Y	Z
A	maintains shape of bacterial cell	protects bacterial cell against desiccation	contains antibiotic resistance genes which may be beneficial to the bacterial cell	serves as the site of protein synthesis
B	controls the passage of substances into and out of the cell	maintains shape of bacterial cell	contains genetic information which is essential to the survival of bacterial cell	serves as the site of translation of mRNA
C	controls the passage of substances into and out of the cell	maintains shape of bacterial cell	contains antibiotic resistance genes which may be beneficial to the bacterial cell	protects bacterial cell against desiccation
D	protects bacterial cell against desiccation	protects bacterial cell from the action of phagocytes	contains genetic information which is essential to the survival of bacterial cell	maintains shape of bacterial cell

18 Which option shows the correct match of processes **W, X, Y and Z** to statements **(1), (2) and (3)**?



(1) NAD is regenerated without the use of the electron transport system.

(2) ATP is synthesised via substrate level phosphorylation.

(3) It can take place under anaerobic conditions.

	(1)	**(2)**	**(3)**
A	Z only	X only	W, X, Z only
B	Z only	X, Y only	W, X, Z only
C	Y, Z only	X only	W, X, Y, Z
D	Y, Z only	X, Y only	W, X, Y, Z

19 Dinitrophenol is a compound that can lodge within the thylakoid membranes of chloroplasts. Its presence provides an alternative route for H^+ ions to diffuse across the thylakoid membranes.

In what way would the Calvin cycle be affected in chloroplasts poisoned with dinitrophenol?

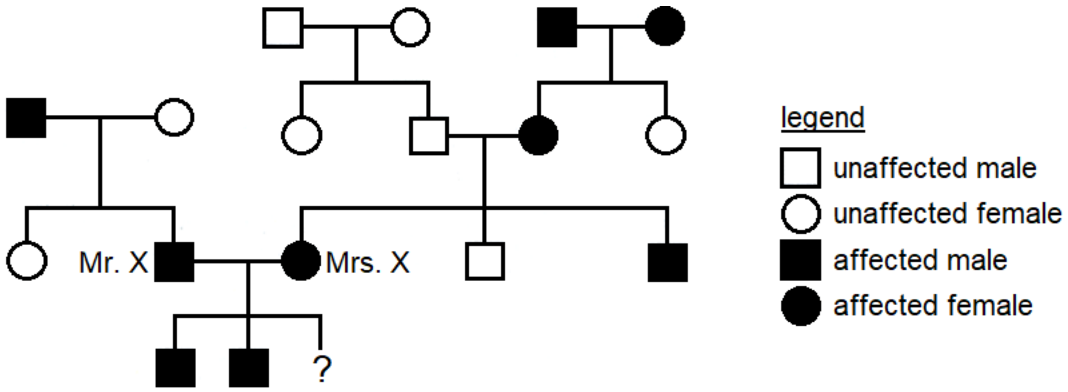
A No effect since Calvin cycle is an enzyme-controlled process.

B The rate of Calvin cycle would increase as pH in the stroma decreases.

C The rate of Calvin cycle would decrease with the accumulation of glycerate-3-phosphate.

D The rate of Calvin cycle would decrease with the accumulation of glyceraldehyde–3-phosphate.

20 The family tree below shows the inheritance of a heart disease due to hypercholesterolaemia.



Mrs. X is expecting a third child. If the child is a son, what is the percentage probability that he will be unaffected?

A 0 %

B 12.5 %

C 25 %

D 50 %

21 Coat colour in horse has three possible phenotypes, grey, black and chestnut, which is due to the interaction of two genes.

Gene ***G/g*** has two alleles: ***G*** resulting in grey coat and ***g*** resulting in non-grey coat.

Gene ***E/e*** has two alleles: ***E*** resulting in black coat and ***e*** resulting in chestnut coat.

The following crosses were carried out.

cross	parental phenotypes	offspring phenotypes
1	grey x grey	grey, black, chestnut
2	black x black	black
3	grey x black	grey, black, chestnut

Which row shows the genotypes of the parents?

	1	2	3
A	GgEe x GgEE	ggEE x ggEE	GgEe x ggEe
B	GgEe x Ggee	ggEe x ggEe	GGEe x ggEe
C	GgEe x GgEe	ggEE x ggEE	GGEe x ggEE
D	GgEe x GgEe	ggEE x ggEe	GgEe x ggEe

22 Two pure-breeding varieties of plants, one producing short leaves and one producing long leaves, were crossed. The resultant seeds were planted in two different locations, and the lengths of the leaves were measured.

	Location A	**Location B**
Number of leaves measured	5	5
Mean length / cm	12.1	6.9
Standard deviation	1	1

The formula used for *t*-test is:

$$t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

$\bar{x}$ = mean of samples
s = standard deviation
n = number of samples

Degree of freedom	**Probability**				
	0.20	**0.10**	**0.05**	**0.02**	**0.01**
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169

Which conclusion is correct?

	t value	**Conclusion**
A	6.544	Differences in lengths of leaves due to different genotypes
B	8.222	Differences in lengths of leaves due to random chance
C	8.222	Differences in lengths of leaves due to effects of different environments
D	6.544	Differences in lengths of leaves due to effects of different environments

23 Which of the following correctly matches the step involved in gel electrophoresis to its purpose?

	Step	**Purpose**
A	Adding buffer solution into electrophoresis gel box	To separate the two complementary strands of double-stranded DNA fragments
B	Adding loading dye	To bind to all DNA fragments to monitor progress of electrophoresis
C	Electrophoresis	To separate DNA fragments based on the amount of negative charges they have
D	Soaking gel in ethidium bromide	To bind to all DNA fragments for visualisation of the DNA bands

24 *Dolly The Sheep* was the first mammal to be cloned from an adult cell. The success of this cloning experiment is consistent with the view that

A differentiated cells retain all the genes of the zygote.

B genes are lost during differentiation.

C the differentiated state is normally very unstable.

D cells can be easily reprogrammed to differentiate and develop into another kind of cell.

25 Which of the following statements about insulin receptor are **false**?

1. It dimerises when insulin molecules bind to each of the 2 receptor subunits.
2. It exhibits enzymatic activity.
3. It has a 7-pass transmembrane domain.
4. It is secreted by β-cells of islets of Langerhans.

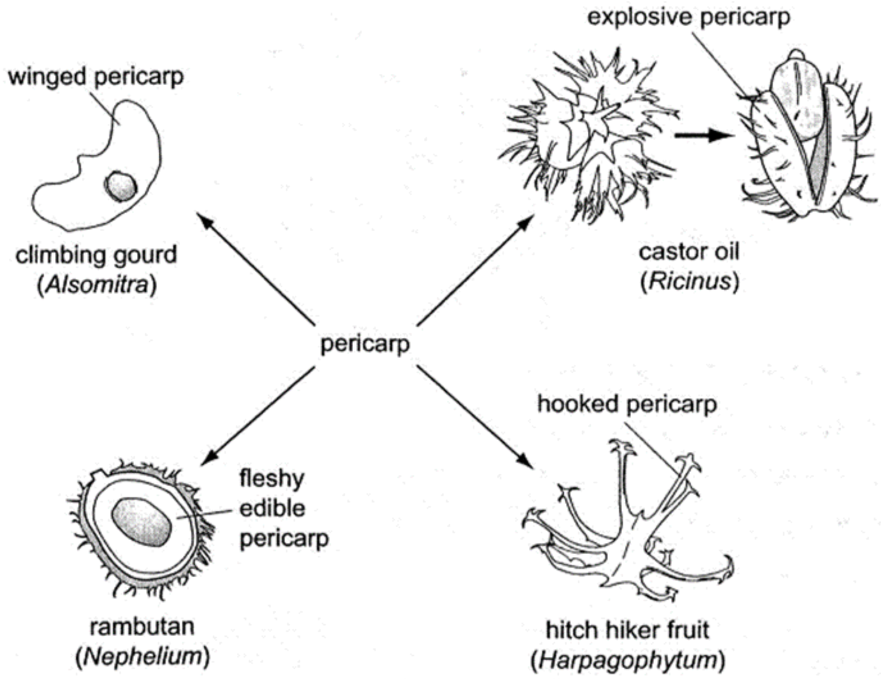
A 1 and 4

B 2 and 3

C 1, 3 and 4

D 2, 3 and 4

26 The diagram illustrates variation in the pericarp (fruit wall) for a variety of methods used in seed dispersal.



What do these examples illustrate?

A The adaptive radiation of analogous structures showing convergent evolution.

B The adaptive radiation of analogous structures showing divergent evolution.

C The adaptive radiation of homologous structures showing convergent evolution.

D The adaptive radiation of homologous structures showing divergent evolution.

27 The α, β and γ globin chains of human, chimpanzee, gorilla and gibbon haemoglobin were analysed. The number of differences in the amino acid sequences in comparison with the human molecules is shown in the table.

species	number of differences in amino acid sequence from human molecules		
	α globin	β globin	γ globin
chimpanzee	0	0	1
gorilla	1	1	1
gibbon	3	3	2

What do the differences suggest?

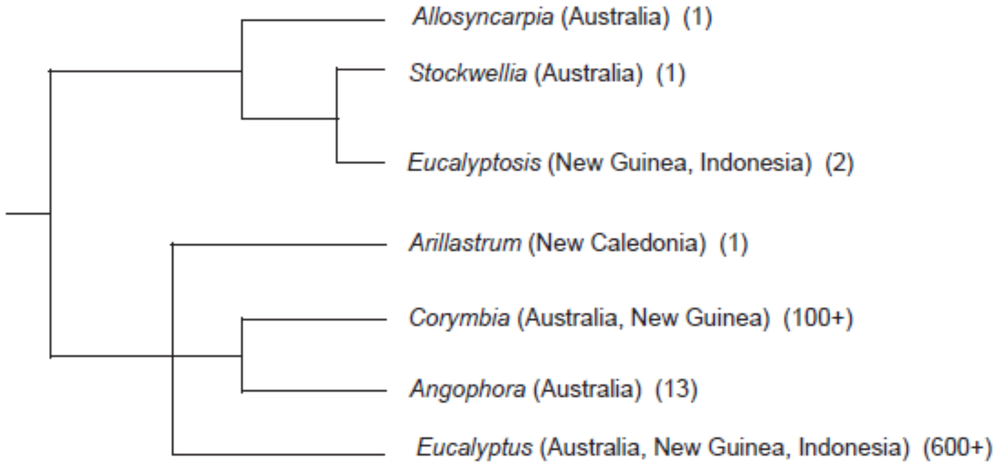
A Humans and gorillas shared a common ancestor more recently than humans and gibbons.

B Humans and gibbons shared a common ancestor more recently that humans and chimpanzees.

C Humans evolved from chimpanzees.

D Humans and gorillas do not share a common ancestor.

28 A proposed phylogeny for the seven genera of plants is shown in the diagram below, along with the countries in which they are found.



Key Numbers in brackets ( ) refer to the numbers of species in the genus.

It would be reasonable to conclude that

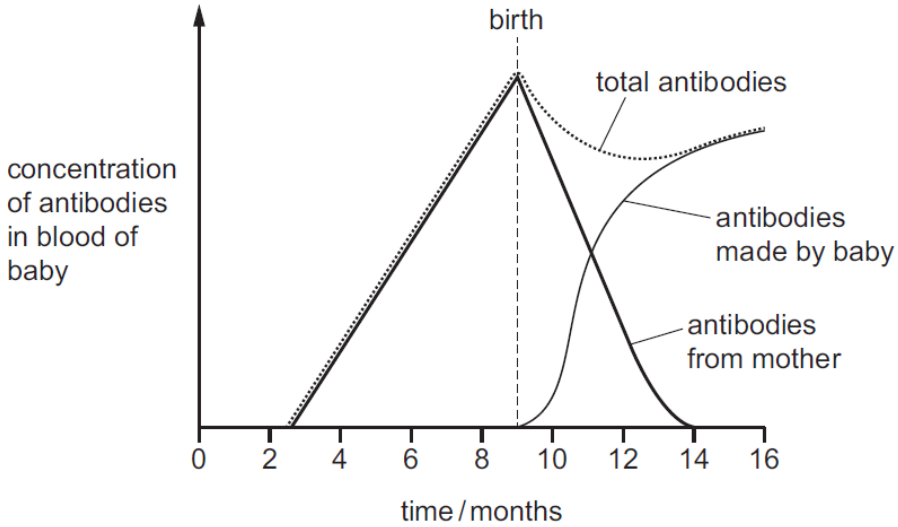
A DNA sequences in *Eucalyptosis* would be more similar to those in *Allosyncarpia* than to those in *Stockwellia*.

B speciation in *Eucalyptus* was assisted by different selection pressures.

C the greater the number of species in a genus, the younger the genus.

D the genus that evolved most recently was *Angophora*.

29 The graph shows the changes that occur in the concentration of antibodies in the blood of a baby before birth and during the first few months after birth.



Which description about the changes in immunity during the first few months after birth is correct?

A active artificial immunity decreases, active natural immunity increases

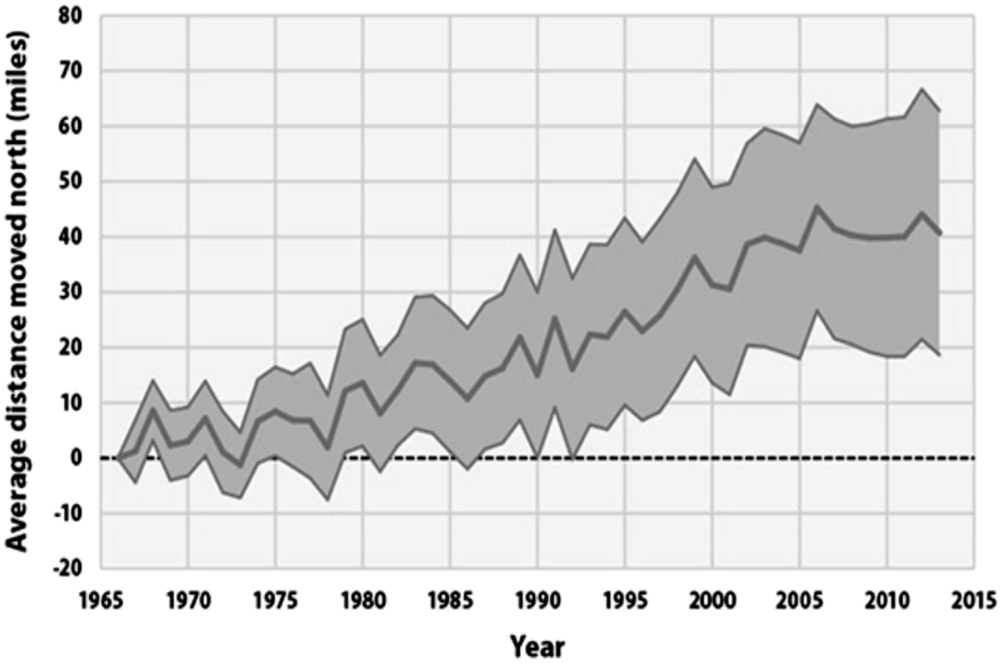
B active natural immunity decreases, active artificial immunity increases

C passive artificial immunity decreases, active natural immunity increases

D passive natural immunity decreases, active natural immunity increases

30 Warmer temperatures are forcing birds in pine forests to breed farther north. Many species once found farther south are also expanding their ranges.

The graph below shows the average latitude occupied by 305 bird species in North America during the winters of 1966 to 2013. The shaded band shows the range of latitudes occupied by the birds.



What could explain the observation?

1 Seasonal birds begin their migration earlier, and lay eggs earlier, in response to warming forest climate.

2 Birds are mobile, thus do not need to adapt and can switch their home ranges and habitat to find more suitable breeding grounds.

3 As temperature rises, hardwood forests in the north lose their advantage, and pine forests found in the south now cover the northern region.

4 As temperature rises, birds experience warmer winters that increases their reproductivity, resulting in larger bird populations.

A 2 only

B 2 and 3 only

C 1, 2 and 4 only

D 1, 3 and 4 only